

Application for the Villum Foundation

March 2025

Civil Engineering and the Green Transition in the Built Environment



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A: Summary of application

Civil Engineering and the Green Transition in the Built Environment (working title, abbreviated CEBE) aim to transform civil engineering practices to create a built environment that aligns with planetary boundaries. Led by Aalborg University, Technical University of Denmark, Aarhus University, and University of Southern Denmark, it focuses on developing sustainable methods, materials, and technologies to reduce greenhouse gas emissions and raw-material use.

Vision: CEBE envisions a future built environment in harmony with planetary boundaries.

Expected Outcomes: CEBE will bring about a disruption by fundamentally transforming civil engineering practices creating real environmental impact. It will position Denmark as a leader in the green transition of the built environment. It will foster scholarly renewal, international collaboration, and excellent research-based education, resulting in high-impact scientific publications and industry collaborations.

Sectorial Approach: The programme adopts a sectorial approach, leveraging the strengths of partner universities and fostering cross-university collaboration. This ensures long-term strategic development and creates enduring academic and industrial networks. The programme will facilitate interdisciplinary collaboration, promote networking, and co-fund collaborative research projects.

Research Fields:

1. Assessing and Measuring Sustainability
2. Design for Regeneration, Circularity, and Longevity
3. Low-Carbon and Robust Construction Materials
4. Digitalization and Automation
5. Climate Resilient and Adaptive Infrastructure
6. Sufficiency, Wellbeing, and Energy Efficiency
7. Extending Life of the Built Environment.

Coordination and Collaboration: A dedicated coordination stream will enhance programme cohesion and optimize synergies across research fields. The programme will engage with national and international academic partners, industry stakeholders, and policymakers to ensure impactful research and practical applications.

Impact: CEBE will engage in policy improvements, in swift adoption of sustainable building practices, and continuously coordinate with global initiatives. It aims to foster resilient, long-lasting infrastructure and buildings, significantly reducing the carbon footprint associated with demolition and new construction.

Budget and startdate: The programme has a total budget of 1 billion DKK over ten years. The expected start date is 1 September 2025.

B: The vision for the project

Humanity's resource consumption considerably exceeds the Earth's capacity to regenerate. The resource consumption in the built environment is one of the main factors as it is responsible for 37 % of the world's energy related greenhouse gas emissions, 31 % of the world's material footprint and 35 % of waste generated, leading to considerable impacts on planetary boundaries like climate change, biosphere integrity and land-system change. Barring a step change in the building sector as a whole, the products, processes, and practices currently in use will continue to represent increasing threats to the human and the natural habitat and undermine the very foundations of welfare and democracy in our societies.

Civil Engineering and the Green Transition in the Built Environment (working title, abbreviated CEBE) will bring about a disruption in human societies' ability to handle challenges by fundamentally transforming civil engineering practices through the development of innovative, sustainable methods, materials and technologies reducing GHG emissions and raw-material resource use.

Vision: A future Built Environment in Harmony with Planetary Boundaries

CEBE contributes strongly to realize this vision by a bold and ambitious investment in the research capacity of Danish universities in the field of civil engineering and the built environment with a focus on achieving world class research excellence and encompassing research-based knowledge and education, human resources, research infrastructure, and industry collaboration.

CEBE addresses seven **knowledge gaps**, which hold the key to a swift and wide transition towards a sustainability paradigm in civil engineering. These knowledge gaps frame the seven research fields (RF) forming the backbone of the sectorial approach developed in the present application:

Gap 1: From Climate Impact to Planetary Boundaries. A shift in LCA methods aligned with absolute targets and tailored the built environment is required to ensure absolute sustainability.

Gap 2: Designing for Regeneration, Circularity and Longevity. There is a lack of engineering design methods for regenerative buildings to mitigate climate change and reduce resource use.

Gap 3: New and alternative construction materials. New understandings are needed to enable a science-based approach to the engineering of new regenerative construction materials.

Gap 4: Digitalization of Construction: Deployment of advanced digital tools and automation technologies is essential to enable scalable solutions for resource efficiency and material circularity.

Gap 5: Resilient and Adaptive Infrastructure: There is a lack of knowledge on impact of climate hazards on facilities and infrastructure, and a lack of understanding of feasible solutions concepts.

Gap 6: Sufficiency in the Built Environment: New understandings on sufficiency are required in management, use and transformation of the existing building stock to mitigate climate change.

Gap 7: Life Extension of Buildings and Infrastructures. New knowledge is needed to enable safe life extension and develop innovative retrofitting strategies on a risk-informed basis.

CEBE is developed and implemented by Aalborg University (AAU), Technical University of Denmark (DTU), Aarhus University (AU), and University of Southern Denmark (SDU) with civil engineering departments, working with international industry and academic partners.

C: Expected outcomes

The **anticipated impact** of the programme is to position Denmark to lead the green transition in the built environment through scholarly renewal, international dynamism, and excellent research-based education; and to reduce the innovation gap in the building sector by promoting sustainable products, processes, and solutions with wide societal impact. This impact will result from a comprehensive sectorial investment in research, education, and infrastructure. At the core of the programme, a push in academic excellence will yield high impact scientific publications, driven collaboratively around a new generation of civil engineers. To maximize impact and the potential for breakthrough, the programme will throughout its lifetime work closely with industry: establishing novel research-driven industry collaborations through lasting academic–industry networks, joint workshops and conferences, joint research projects, co-funding opportunities, and technology transfer initiatives.

From a scientific perspective, the programme’s ambition is to reach world-class research capacity and quality by exploiting the synergies of an integrated sectorial approach, which will inspire major scientific contributions, including new methods, solutions and technologies, with wide societal impact. To maximize the synergies of the sectorial approach the programme will throughout its lifetime promote networking and facilitate and co-fund collaborative research projects, and data and infrastructure sharing. The anticipated outcomes include groundbreaking knowledge on LCA, design methods, regenerative materials, new technologies and solutions; a new generation of engineers equipped with critical skillsets to lead the green transition; and a robust pipeline of talents for academia and industry that will continue to benefit Danish civil engineering beyond the programme.

From a wider societal perspective, the technical and societal benefits from the project will engage in improving policy and regulatory standards, a wide and swift adoption of sustainable building practices through lifelong up-/re-skilling and new spinouts and job creation.

There will be continuous coordination with Danish and global initiatives to catalyse the green transition in civil engineering and leverage additional funding.

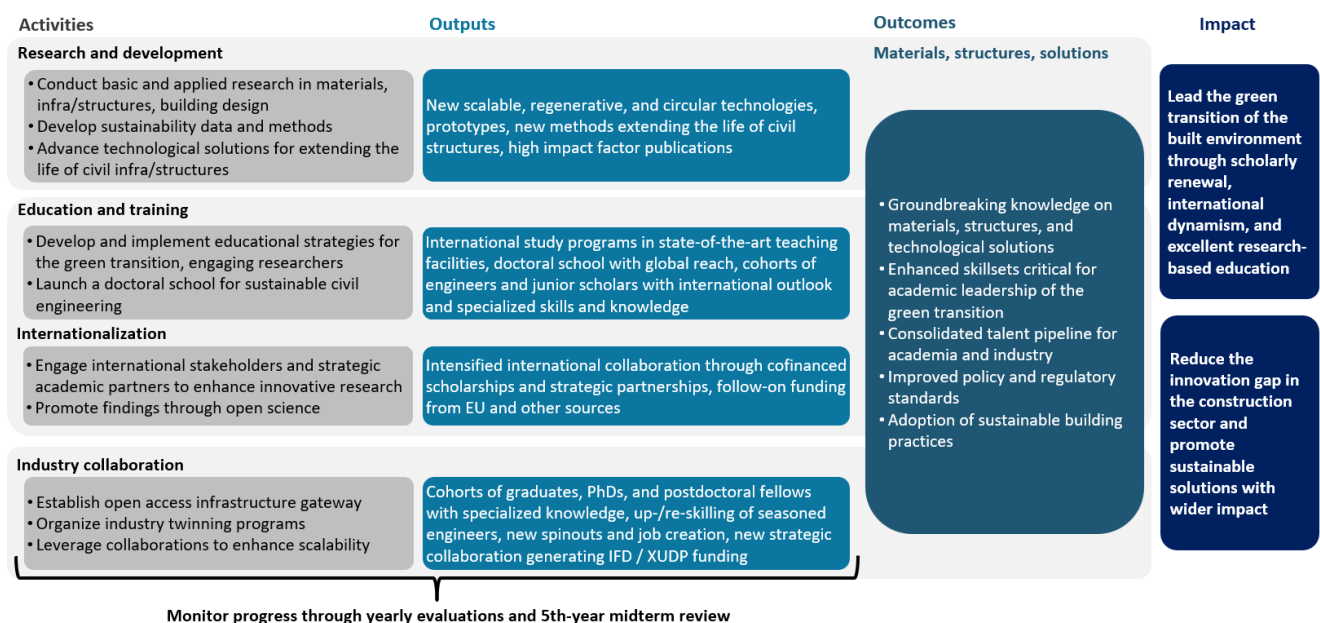


Figure 1. Pathways to impact

D: The sectorial approach

The programme has chosen a sectorial approach to address the very diverse research challenges required to enable a future built environment in harmony with planetary boundaries, see Figure 2.

The sectorial approach ensures a long-term strategic development by leveraging universities' strengths and aligning them under a unified national agenda. The approach exploits both university-specific sectorial expertise and cross-university collaboration to create the critical mass needed to significantly strengthen research capabilities and establish internationally recognized spearhead positions within each of the seven research fields. The sectorial approach will create enduring academic and industrial networks that continue to empower Danish research and education in sustainable civil engineering beyond the life of the project, ensuring that innovations are continuously refined and scaled through sustained international collaboration and follow-on funding opportunities.

The Scientific Coordination Group and the coordination stream will continuously ensure alignment of efforts, sharing of insights, and momentum of each research field toward the project's green-transition objectives. Programme management, activities and open calls will facilitate interdisciplinary collaboration across research teams, leveraging research field synergies and ensuring that early-career researchers are fully integrated into the initiative. This will enable universities to align their expertise, collaborate, and prioritize their efforts in educating the next generation of researchers, in attracting international talent, and in facilitating impactful, research-driven innovation within the built environment fostering scalable solutions that are of high value at the international level.

The selection of the seven research fields is based on the research gaps identified and a collaborative and comprehensive mapping of existing Danish strongholds, supported by the expertise of the Advisory Board. Combined, the research fields cover all the life cycle stages of the built environment from design, construction, operation, life extension to demolition, and they will provide directions for actions of change in engineering practices.

Within the programme, emphasis is put on the research fields which focus on low-carbon construction materials, climate resilience of infrastructures, and extending life and use of the built environment.

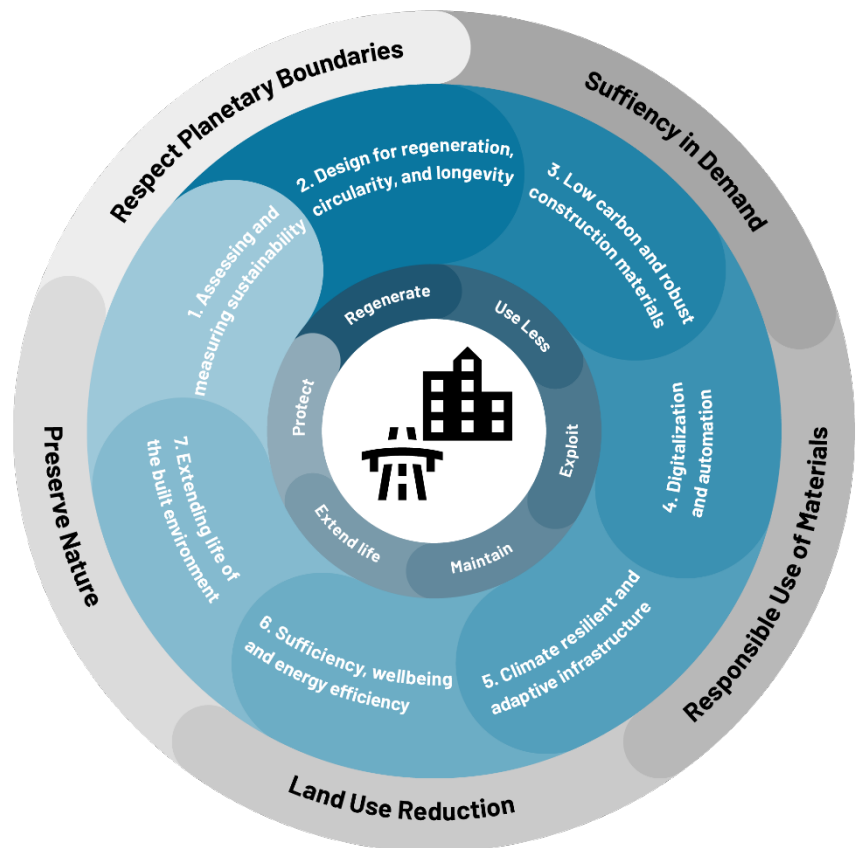


Figure 2. The programme sectorial approach

Response to report from Advisory Board

Research Field (RF)	Response
Coordination Stream	The application includes a number of measures to strengthen cohesion among all researchers. To facilitate collaborative research and sharing of data and infrastructure, the coordination stream will regularly update partners on research achievements and collaboration opportunities through internal communication, online meetings and PhD seminars, in-person research field workshops, and a yearly two-day seminar.
RF1. Assessing and measuring sustainability	The scope of Research Field 1 (RF1) is kept within environmental sustainability and therefore economic and social indicators not yet included. Broadening the scope will be a strong driver in future efforts for attracting additional funding. We aim to include stakeholders from industry in workshops etc. and to have frequent releases of methods and data that can be adopted by industry. An important uptake of methods is also planned within RF2-RF7. Two of four WPs include focus on longevity of built environment and therefore focus on longer service life.
RF2. Design for regeneration, circularity, and longevity	The content has been revised following the advice from the AB who suggested to reduce the number of themes and focus on engineering design of buildings and processes. This RF is now focusing on the development of methods and processes for engineering design of regenerative buildings and developing circular building components that prioritize reuse, recycling, and the integration of biogenic materials, ensuring long-term durability and adaptability in a changing climate.
RF3. Low-carbon and robust construction materials	A major vision for the RF is expressed in the text in relation to global adaptation “We wish to use Denmark as an international showcase as large-scale proof-of-concept for scaling the alternative and new construction materials. We will be the scientific base for an international movement that targets absolute sustainability for materials used in buildings and structures.”
RF4. Digitalization and automation	This research field has been restructured into two areas: "Construction Automation" and "Data-Driven Construction and Operation," reducing work packages to six. WP 4.1 now focuses explicitly on reinforced 3D concrete printing. WP 4.4 and WP 4.5 address the existing built environment through "Reality Capture" and "Digital Twins for Building Management," respectively. Industry partners will co-develop projects post-proof-of-concept, with collaboration aligned to the TRL's of each work package.
RF5. Climate resilient and adaptive infrastructure	The inclusion of industry is clearly stated under "Key stakeholders" and "Plans for research impact". The industry is seen as a key partner in Joint Pilot and Demonstration Programmes (JPDP). Decision processes associated with climate resilience is closely related to risk assessment, which is a WP on in itself, WP5. In WP3 the task, "5.3.4 Decision and management strategies regarding the need to protect, adapt, or abandon", actively includes stakeholders in decision and management strategies.
RF7. Extending the life of the built environment	Overlaps with RF3 have been eliminated. A new lead from AU and a co-lead from DTU have been appointed to strengthen the focus on structural engineering. The ambitious research program contains projects addressing important and tangible problems, such as assessment of non-code compliant and/or corrosion-deteriorated structures. The entire program aims to maximize the service life of the built environment with minimum intervention.

Table 1. Response to report from Advisory Board

E: Sectorial activities

The main research activities are planned and carried out in seven research fields (RF). The four partners have teamed up within each RF to ensure that one or more of the universities involved already play an international role in the RF and has the potential to further develop and create a global spearhead. Each RF is managed by a lead, assisted by co-lead(s) from the other partner universities. The university partners within each RF have developed a joint plan enabling them to align their expertise, collaborate, and prioritize efforts within designated areas of the joint effort.

Research Field	0	1	2	3	4	5	6	7
	Coordination stream	Assessing and measuring sustainability	Design for regeneration, circularity, and longevity	Low-carbon and robust construction materials	Digitalization and automation	Climate resilient and adaptive infrastructure	Sufficiency, wellbeing and energy efficiency	Extending life of the built environment
AAU	Lead	Lead		Co-lead		Co-lead	Lead	
DTU	Co-lead	Co-lead		Lead	Co-lead	Lead	Co-lead	Co-lead
AU	Co-lead		Lead			Co-lead	Co-lead	Lead
SDU	Co-lead	Co-lead	Co-lead		Lead		Co-lead	Co-lead

Table 2. The role of each partner university in the research fields

RF 1 focuses on assessing and measuring sustainability. To measure sustainability and map the built environment against the planetary boundaries, it is essential to highlight high-impact strategies for the green transition. Each institution brings decades of experience to the partnership. AAU brings expertise in sustainability assessment and LCA tools. SDU specializes in industrial ecology and dynamic LCA. DTU focuses on impact assessment and absolute sustainability research.

RF 2 develops new design methodologies encompassing circularity, longevity, and regeneration. Developing new design methodologies and ensuring their implementation is essential to achieve industrial and societal impact. AU and SDU have a strong track record in integrating digital predictive performance assessment that enables building designers to assess social, economic, and environmental consequences of design options in the early stages of the building design process.

RF 3 targets the development of low-carbon and robust construction materials. New materials, new material technologies, and materials with less use of virgin materials are essential for decarbonizing the industry. Both DTU and AAU have strong research competencies in the area. DTU conducts applied research within reuse and recycling on solid foundation of basic material science. AAU adopts a multidisciplinary approach that combines experimental testing, theory, and probabilistic methods.

RF 4 focuses on digitalisation and automation technologies as they are essential for achieving the ambitious global sustainability goals. SDU brings expertise in digital fabrication, construction 3D printing. DTU complements with research experience in digital construction twins, data engineering interoperability, 3D printing and analytics, and machine learning.

RF 5 focuses on the climate change induced risks to infrastructures and coastal areas, which require more research in climate-resilient and adaptive infrastructures. DTU contributes with extensive experimental facilities and advanced hydrodynamics and hydraulic modelling skills. AAU has expertise in urban stormwater drainage, wastewater handling, and urban hydrology. AU supplements with ocean monitoring, advanced machine learning, and comprehensive capabilities in geotechnics.

RF 6 develops a sufficiency approach to reach sustainable and healthy built environments requiring research that integrates management, use of buildings, comfort norms and energy performance. Together, the four universities embrace expertise in all relevant aspects of this research field.

RF 7 adapts the strategy of extending the built environment's life as a prominent decarbonization approach. AU, DTU, and SDU bring expertise in structural assessment, system identification and health monitoring, and structural engineering of existing bridges and building structures.

RF, Work Packages	Main outputs/Milestones
1. Assessing and measuring sustainability	
1.1 Planetary boundaries and their application to LCAs	Development of novel methods for measuring the built environment within planetary boundaries. Uptake and integration of measurement methods in the construction industry. Creation of robust methods to support political targets (net zero, planetary, regenerative). Adoption of methods for measuring policy progress.
1.2 Expanded system boundaries	
1.3 Embracing the temporal dimension in LCAs	
1.4 Absolute environmental targets and claims	
2. Design for regeneration, circularity, and longevity	
2.1. Design of regenerative buildings	Regenerative building design paradigm and methods. Design methods for circular solutions. Regenerative strategies for existing building stock. Integration of regenerative principles in urban building. Industry adoption of regenerative construction. Policy recommendations enabling regenerative construction.
2.2 Design for circularity and longevity	
2.3 Regenerative construction processes	
3. Low-carbon and robust construction materials	
3.1 New theoretic foundation	Model linking parameters of constitutive material models to physical and chemical properties across scales. Reliable assessments for material reuse acceptance. Methodology of secondary resources to replace natural resources in traditional materials. New materials based on fundamental knowledge respecting planetary boundaries. A decision-making framework supporting absolute pathways.
3.2 Reuse of materials	
3.3 Alternative materials with secondary resources	
3.4 Development of new materials	
3.5 Implementation in the building industry	
4. Digitalization and automation	
4.1 Advanced Low-Carbon Prefabrication	Demonstration of 1) automated, reinforced 3D concrete printing low-carbon building elements; 2) autonomous robotic assembly, disassembly, and re-assembly; 3) autonomous robotic deconstruction; 4) a multi-scalar digital twin system, from component to city scale. Real-time tool for extracting geometry, semantics, and materials information from sensordata in buildings and construction.
4.2 Adaptive Prefabricated Engineered Timber Systems	
4.3 Automated deconstruction and material extraction	
4.4 Data-driven interoperability and decision-making	
4.5 Digital Twins for smart construction and management	
4.6 Intelligent Reality Capture	
5. Climate resilient and adaptive infrastructure	
5.1 Acquisition, management, and use of data	Extensive field and laboratory data to support use of advanced models and expert systems in decision-making for adaptation, protection and abandonment strategies. Catalogue of traditional hard (structures) and soft (land management) approaches to adapt and protect infrastructure. Projection of climate change impact scenarios as basis for detailed risk assessments.
5.2 Multiphysics modelling of infrastructure	
5.3 Adapting the coastal and urban zone	
5.4 Adapting civil infrastructure	
5.5 Projecting risk of civil infrastructure	
6. Sufficiency, wellbeing and energy efficiency	
6.1 Building related sufficiency	An integrated model for building related sufficiency. Updated comfort models to facilitate building construction, renovation, and technology development. Tested cases of sufficiency guided construction methods. Pilot projects with industry in real-life settings on smart building operation and management. Sufficiency based guideline for smart grid integration.
6.2 Updated comfort models	
6.3 Sufficiency-guided construction and technology	
6.4 Smart building operation and management	
6.5 Buildings demand response and grid integration	
7. Extending life of the built environment	
7.1 Assessing the capacity and residual life	Quantified structural deterioration using laboratory experiments, structural health monitoring, and limit analysis. Validated residual load capacity modeling, fatigue lifetime predictions, and AI-driven inspection for structural deterioration. Sustainable strengthening and repair methods. Lifetime extension and retrofitting scenarios with cost-optimized reliability targets.
7.2 Predicting future use and load scenarios	
7.3 Engineering solutions for life extension	
7.4 Riskbased decision support systems for life extension	

Table 3. Work packages in the research fields

Although each of the research fields focuses on individual knowledge gaps, the ability to meet the vision of the programme and the expected outcomes and impact, ultimately depends on collaborative efforts and extensive exchange and exploitation of research outputs.

The outputs of RF1 will inform and support all other research fields in evaluating and target setting a "Built Environment within planetary boundaries". RF1 will develop a compass for evaluating and benchmarking sustainability, which will be applied throughout the programme. The compass will also be shared with industry partners for integration into sustainability practice. RF2 will further integrate new materials developed in RF3 in the design of circular and regenerative components and buildings. Research in construction processes and management in RF2 is coordinated with the research activities in digital frameworks and automation technologies for construction and deconstruction in RF4. Research in engineering design methods for regenerative building is coordinated with findings from the research in sufficiency and building operation in RF6. Material innovations in RF3 will support low-carbon materialisation processes, helping to improve construction automation in RF4 and essential for decarbonizing climate resilient infrastructure in RF5. The development of digital twins in RF4 will provide a strong foundation for research into sufficiency and energy efficiency, bridging construction and operations. Retrofitting strategies and risk assessment tools from RF7 will offer useful insights for construction and deconstruction automation in RF4, while new ideas about sufficiency and comfort from RF6 and climate adaption from RF5 will create a new basis for development of retrofitting strategies in RF7. Given these interdependencies, the coordination stream will foster collaboration between research fields, share outputs, and exploit synergies.

Table 4 shows the number of person years (PY) per research field and partner supported by CEBE. The total number of new hires related to sectorial activities will be higher during the 10-year period as several of the PhD and Postdoc positions are financially supported by other project activities. The partners will also be co-financing several of the tenure track assistant professor (TT) positions.

RF		1	2	3	4	5	6	7	Total
Partner	Position	Assessing and measuring sustainability	Design for regeneration, circularity, and longevity	Low-carbon and robust construction materials	Digitalization and automation	Climate resilient and adaptive infrastructure	Sufficiency, wellbeing and energy efficiency	Extending life of the built environment	
AAU	PhD	9,0		34,5		18,0	9,0		70,5
	PD	7,8		14,0		15,2	0,0		37,0
	TT	8,0		17,5		5,0	8,4		38,9
AU	PhD		21,0			24,0	12,0	22,5	79,5
	PD		8,6			12,0	3,8	7,0	31,4
	TT		0,0			5,1	0,0	10,0	15,1
DTU	PhD	9,0		20,7	12,8	15,0	5,3	21,0	83,7
	PD	5,1		14,0	9,7	11,0	3,5	5,1	48,4
	TT	4,3		23,3	86,3	5,0	4,8	5,0	48,6
SDU	PhD	15,0	15,8		20,3		6,0	30,0	87,1
	PD	5,5	7,0		6,6		6,0	10,0	35,1
	TT	3,1	9,2		8,8		6,0	5,0	32,1
Total	PhD	33,0	36,8	55,2	33,1	57,0	32,3	73,5	320,8
	PD	18,4	15,6	28,0	16,3	38,2	13,3	22,1	151,9
	TT	15,4	9,2	40,8	15,1	15,1	19,2	20,0	134,7

Table 4. Person years per research field and partner

On top of this, several open call recruitment and fellowship programmes are integrated to provide renewal and inspiration through recruitment of new international talents and ensure the integration of new ideas from national and international academic and industry partners. This will result in attracting 5-8 new international research talents as well as 50 PhD's and 25 postdocs.

This combined investment will place the project partners in a strong position to attract follow-on funding for excellent research (e.g., ERC, DFF) and for research and innovation activities with industry (e.g., IFD, EU, XUDP).

The expected start date for CEBE is 1 September 2025.

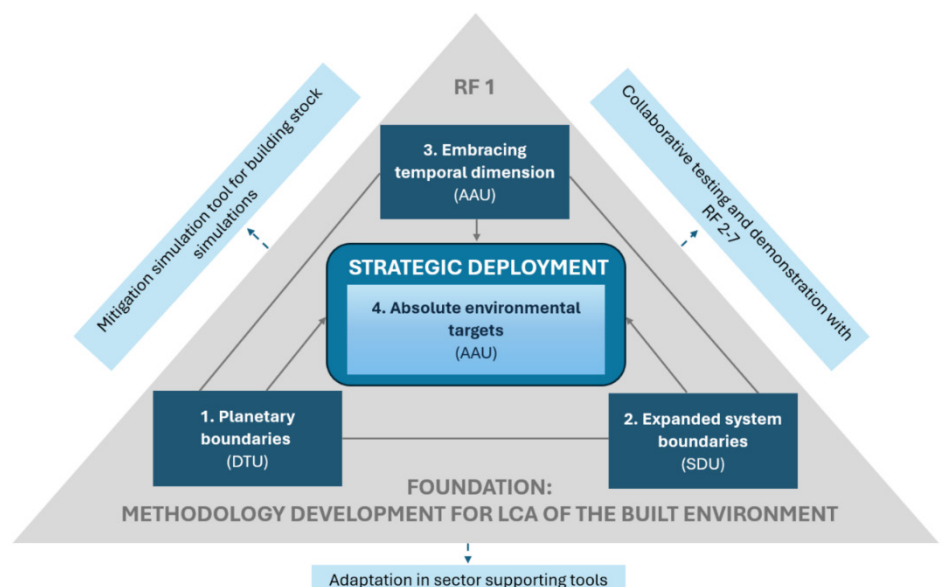
Research Field 1: Assessing and measuring sustainability

Key obstacles to robust environmental sustainability assessment (ESA) for the built environment (BE) currently include reliance on static models in a dynamic world, failing to capture complexities like time-dependent burden-shifting between systems and across environmental impacts. Excessive data demands and outdated/lacking information hinder accurate ESA, while industry resistance delays adoption of new practices. The goal is to refine assessment methods that enable climate change mitigation and environmental sustainability. RF1 aims to develop solutions for (1) guiding the construction industry to meet new green targets, and (2) supporting sustainability goals with dynamic models for both historical and future assessments. This research will identify changes in material types/material flows and methods to promote and enable a climate-neutral and sustainable BE. RF1 will create methodological foundation that other RF's (2-7) can use in evaluating their impact towards transition of the BE within planetary boundaries. RF1 outcome will be shared with industry partners for integration into sustainability practice and with universities for educational resources.

Plan for activities

RF1 will enhance ESA decision support foundation for the BE by addressing key obstacles and unresolved methodological gaps. It will develop robust LCA methods based on absolute environmental targets, including net-zero, planetary boundaries, and regenerative measures (WP1, WP4).

RF1 will refine system boundary definitions and investigate their influence on sustainability evaluations, with a focus on dynamic, prospective models incorporating improved temporal dimensions (WP2, WP3). The foundation will be developed and tested in WP1-3, leading to a comprehensive approach for aligning with absolute green targets (WP4). This methodology will enable other RFs (2-7) to quantify sustainability implications and support strategic deployment in civil engineering, preventing greenwashing and ensuring accurate, actionable assessments. We propose three enablers to amplify the impact in all RFs and the building industry.



WP1 develops a methodology for assessing absolute sustainability impacts on biodiversity, resource use, and chemical emissions in the life cycle of BE. Focuses on characterization modelling of resources, ecotoxicity from chemicals, and regionally differentiated life cycle boundaries.

WP2 compares the impact of simplified vs. expanded system boundaries (SBs) in LCAs for BE. Illustrates how different SBs affect relative/absolute sustainability perceptions. Explores incorporating expanded SBs into LCA practices for enhanced sustainability decision support.

WP3 focuses on dynamic assessment methods for reducing impacts over time without burden shifting. Addresses temporal emission quantifications and future changes in environmental and socioeconomic conditions. Covers long service-life spans of buildings and infrastructure.

WP4 defines absolute environmental impact targets (e.g., climate, biodiversity, resource use) for BE. Targets may include net-zero, planetary, restorative, or regenerative goals. Provides guidance to ensure accurate claims and achievement based on WP1-3 methodologies.

WP	RF1: Assessing and measuring sustainability	Output/milestones
1.1	Planetary boundaries and their application to LCAs of the BE	Lead: DTU. Co-lead: AAU, SDU
1.1.1	Scaling impacts to BE	First version for PB-LCA for the BE with biodiversity impact to be implemented outside RF1 in Q4-26 (MS1a). Updated every third year: MS1b in Q4-29 with resources, energy and chemicals + updated version of biodiversity; MS1c in Q4-32 with biomaterial and temporal perspective in PB.
1.1.2	Characterization modelling (a. resources, b. ecotoxicity/chemicals)	
1.1.3	PB for: a. energy, b. biomaterial consumption	
1.2	Expanded system boundaries in assessments of the BE	Lead: SDU. Co-lead: AAU, DTU
1.2.1	Implications of applying simplified and expanded system boundaries (SB)	MS2a overview of result differences obtained when applying different SB in BE: MS2b proposal for how expanded system boundaries can be applied for assessing BE: MS2c validation of the expanded system boundary approach for BE and its applicability for assessing absolute targets (in-put to WP1,4).
1.2.2	Methods for incorporation of expanded system boundaries in LCA	
1.3	Embracing the temporal dimension in LCAs of the BE	Lead: AAU. Co-lead: SDU, AAU
1.3.1	Model for dynamics in the use throughout the life cycle of buildings	MS3a New approach for understanding the life cycle of the dynamic use of buildings improving absolute metrics for buildings, input into WP1,4. MS3b New dynamic model and data for use in updating approaches and models in WP1,4. MS3c New dynamic multi-scenario model for MFA and LCA that will improve models within WP1,2,4.
1.3.2	Dynamic model for long term projects incl. prospective LCA & time dependent uptake/ release	
1.3.3	Dynamic multi-scenario model for changes in environmental boundary and socioeconomic conditions for BE with long service-life spans	
1.4	Absolute environmental targets and claims in the BE	Lead: AAU. Co-lead: DTU, SDU
1.4.1	Methods for absolute targets and scaling impacts to buildings and BE	MS4a Approach tested and implemented in industry practice and policy for climate neutral buildings (Q3-26), climate neutral BE (Q1-27), BE within PB (Q3-27), regenerative BE (Q1-28). These will be updated every third year (MS4b Q1-31, MS4c Q1-34).
1.4.2	Absolute targets for climate impacts, other env. impacts and resources	

Key stakeholders

RF1 aims to include the following through strategic cooperation, workshops, dissemination of data and methods: Consultants and industry that will apply new methods and integrate more ambitious SEA in their practices. National and municipal authorities and certification organizations that adopt higher ambition levels. Data providers and standardization bodies that will improve data quality. LCA tool developers and users. International academia and RF2-7 are important stakeholders.

Plans for research impact

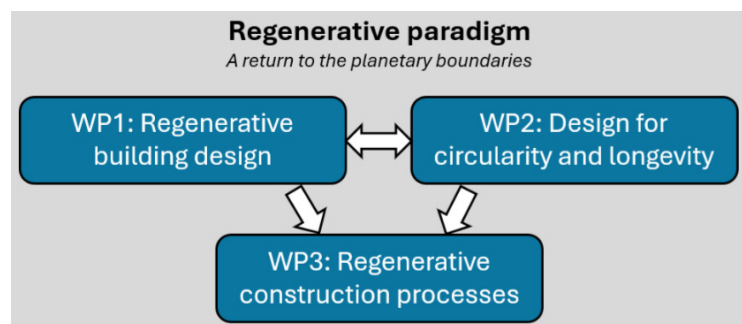
RF1 will change the way we apply LCA for the BE, generating scientific impact through high-level peer-reviewed publications. We will develop and demonstrate methods for assessing BE performance within absolute targets. To ensure societal impact, RF1 aims for early adoption by other RFs (2-7),

industry partners, and global academia. Iterative publication of refined methods and data will be essential. RF1 will deliver assessment methods that support national and international political commitments across all key environmental impact indicators, facilitating early policy implementation.

Research Field 2: Design for regeneration, circularity, and longevity

The environmental load from the built environment must be reduced to a level where the regenerative properties of Earth's ecosystems become able to restore the planetary boundaries. The ambition of RF2 is to establish the scientific foundation for a paradigm based on regeneration (Mang & Haggard, 2016) when applying civil engineering for the design of new buildings.

The goal of the research is to define methods and processes for regenerative engineering design of buildings where circular components and systems based on reuse and recycling of existing materials and new biogenic materials are integrated to ensure longevity of solutions in a changing climate. RF2 will lead research efforts and carry out collaborations with international universities as well as the national and international building industry. The expected outcome is a paradigm for design of new buildings that leads to buildings with a regenerative environmental impact.



Plan for activities

The research is organized in three work packages running in parallel throughout the ten-year project period. WP1 is *Regenerative building design* where the regenerative paradigm is the foundation for the development of new design methods and processes that enable the building industry to design long-lasting circular buildings ready for future climates. WP2 is *Design for circularity and longevity* where focus is on how to engineer long-lasting building components and systems based on biogenic materials, reuse and upcycling of existing products that can be recirculated repeatedly.

WP3 is *Regenerative construction processes* where new construction processes for regenerative construction and O&M systems are designed. The WPs are complementary as design of components and systems (WP2) and design of buildings (WP1) affects one another, while construction processes and O&M systems (WP3) must be designed to realise the design intentions.

The research activities within this field aim to develop and test novel approaches to regenerative, circular, and long-lasting building design. The focus is on advancing scientific knowledge and translating it into practical applications through experimental studies, case-based research, and collaboration with industry partners. The work will integrate digital design methods, material innovations, and systemic approaches to ensure adaptability and resilience in the built environment.

Through interdisciplinary efforts, methods that enable the transition toward buildings and construction processes that actively restore environmental balance while meeting societal and functional needs will be established. The first four years also include the establishment and operation of a Regenerative Design Lab for civil engineering students with facilities for testing and developing novel design methods, building components and systems.

WP	RF2: Design for regeneration, circularity, and longevity	Output/milestones
2.1	Design of regenerative buildings	Lead: AU. Co-lead: SDU
2.1.1	Building design methods in a regenerative paradigm	Explicit regenerative design paradigm (Q4-26). Regenerative building design methods (Q1-28). Practical building examples (Q1-30)
2.1.2	Climate-change ready thermal design of buildings	Method for designing, redesigning and preparing buildings to minimize the risk of overheating due to climate change (Q1-28)
2.1.3	Regenerative design of neighborhoods	Design method for transforming buildings into regenerative ecosystems through bioreceptive design of urban facades (Q2-28)
2.2	Design for circularity and longevity	Lead: AU. Co-lead: SDU
2.2.1	Designing regenerative facades	Design methods for creating circular solutions based on reused and biogenic materials (Q1-28)
2.2.2	Long-lasting biogenic building components	Design methods for ensuring the longevity of biobased building components and systems in a changing climate (Q1-28)
2.2.3	The circular window	Design-for-disassembly of windows using existing panes, novel transparent layering, and innovative framing (Q1-31)
2.2.4	Upcycling waste into structural components	Stock-driven circular design and fabrication of structures for buildings (Q4-29)
2.3	Regenerative construction processes	Lead: SDU. Co-lead: AU
2.3.1	Design of regenerative construction processes	Construction processes that are aligned with the regenerative paradigm (Q4-29)
2.3.2	Integrated concept for circular design and construction	Concepts that merge regeneration, longevity and circularity across project stages (Q2-30)
2.3.3	Construction management systems for regeneration	Methods for a regenerative environmental impact from the built environment through innovative construction planning systems (Q4-29)

Key stakeholders

Research collaboration with various types of stakeholders is anticipated. This includes research projects in partnerships with international research institutions, using case studies relevant to municipalities, social housing associations, and private building owners. Engagement with industry partners will focus on co-creating circular solutions and design methods that can be implemented at scale, ensuring real-world applicability. Collaboration with private companies will drive innovation in material reuse, biogenic components, and regenerative construction techniques. Additionally, partnerships with Non-Governmental Organizations (NGOs) will help push the regenerative building paradigm forward by advocating for policy changes and promoting sustainable practices across the sector. Active dialogue with regulatory bodies and certification organizations will also play a role in shaping future frameworks for regenerative and circular building design.

Plans for research impact

The scientific milestones of the research activities within the first four years are 1) a regenerative building design paradigm, 2) functional concepts for circular building components and systems, and 3) regenerative construction processes. To maximize impact, findings will be shared through high-impact journal publications and conference proceedings, increasing visibility within the academic and professional communities. Research outcomes will also be integrated into industry practice through collaborations with R&D departments, professional networks, and public-private partnerships. Regular communication on knowledge-sharing platforms and open-access databases will help promote the adoption of regenerative design principles. Long-term partnerships with academia, policymakers, and industry stakeholders will further advance guidelines and standards to accelerate the transition to a regenerative built environment.

Research Field 3: Low-carbon and robust construction materials

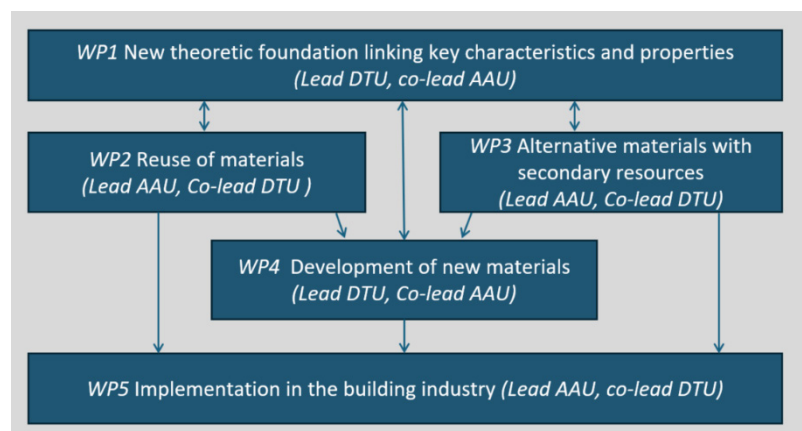
RF3 will enable *science-based development, engineering, and implementation of alternative robust construction materials based on stringent and fundamental principles while respecting planetary boundaries*. Today's most used construction materials have a vast environmental burden, highlighting the need for sustainable replacements. Balancing urgency and demand for sustainable materials is a massive challenge. The research strategy has two tracks. A short-term strategy (5 years) for significantly improving the environmental impact of known materials through alternatives based on reuse and recycling without compromising performance. A long-term strategy (5-10 years) provides science-based tools to develop and document new low-carbon, resource-efficient, and robust materials. *We will work simultaneously at all levels between science-based developments and implementation to meet the urgency of documenting alternative and new structural materials for the building industry.*

Plan for activities

The research lies in the tension field between basic research (WP1) and implementation (WP5). All 5 WPs will be active over 10 years. The mutual weight will change following short- and long-term strategies.

A strong focus throughout the project will be on fundamentals (WP1) to enable the ambitious shift from today's material use to tomorrow's sustainable use.

The urgent need for more environmentally sound materials necessitates faster implementation of alternatives. This calls for direct reuse of materials and components (WP2) and the development of alternatives to known materials and components based on recycling (WP3). The strategies are researched worldwide but are barely implemented, as international research is generally fragmented and nonsystematic. In (WP4) we develop new materials based on the framework in (WP1) and generic findings in (WP2) and (WP3), with the long-term goal to develop new materials that we do not even know today produced while respecting planetary boundaries. In (WP5) faster and safer implementation will be targeted through the systemization of research findings and collaboration with stakeholders in the sector.



WP1: Predicting the behavior of construction materials by unraveling the microstructural origin of macroscopic behavior will lead to a paradigm shift from current practice based on experience. This new fundamental base will support the development of materials and test methodologies. We will develop a central model linking parameters of constitutive material models to physical and chemical properties across scales, establishing correlations through statistical models and AI. The new theoretical frame will be developed and calibrated using known materials. These principles will support the development of robust and documented materials in WPs 2-4.

WP2: Non-destructive testing of materials and components for reuse for documenting properties in environmental classes beyond X0. The goal is to establish reliable assessments and criteria for

material acceptance to maximize the reuse potential. A major question is how the deterioration continues when the component is moved to another environmental class.

WP3: Improving the environmental impact of known materials without compromising performance is of utmost importance. The use of secondary resources (biological or mineral) is one important direction of the research, and a strong focus will be on durability and potential toxic compounds.

WP4: Develop and document new low-carbon, resource-efficient, and robust construction materials based on the science and framework from WP1. The materials will be targeted towards their environmental conditions. Statistical tools, stochastic modeling, and AI will be used. Focus will be on parameters important for industrializing production. Knowledge from WP2 and WP3 will support including multiple life thinking in new materials for future circular use.

WP5: Bridging the gaps between research and industrial adoption is in focus. Establishing a reliability framework for introducing new and alternative materials is key. Methodologies to document the properties of new materials are implemented via guidelines and standards simultaneously with developments so they enter into force as soon as possible. Industrialization is crucial for accelerating and scaling, and transition theory is used to frame the sectorial innovation and industrialization processes. A decision-making framework to support absolute sustainable pathways for construction materials value chains will be developed.

WP	RF3: Low-carbon and robust construction materials	Output/milestones
3.1	New theoretic foundation linking key characteristics and properties	Lead: DTU. Co-lead: AAU
3.1.1	Central model linking the parameters of constitutive material models to physical and chemical properties across scales supported by AI, hybrid testing, and digital twins.	(M1) Use cases for AI of constitutive model parameters defined for digital twin framework.
3.1.2	Experimental methodology and generic model for forecasting degradation and property changes of sustainable materials	(M2) Test methodology designed combining traditional and new, advanced experimental methods (M3) First validation of central model with most known materials
3.2	Reuse of materials	Lead: AAU. Co-lead: DTU
3.2.1	Life expectancy of structural components for reuse. Testing and modeling.	(M1) Database and data-driven models as base for continuously refining non-destructive experimental protocols for reused elements
3.2.2	Development of an experimental design framework for optimizing component selection and ensures safe reuse	(M2) The framework for sampling and testing of components for reuse is demonstrated numerically and developed for reused components.
3.3	Alternative materials with secondary resources	Lead: AAU. Co-lead: DTU
3.3.1	Performance-based design framework based on selected materials for combining secondary resources into alternative materials and components	(M1) Applied and tested the framework on selected materials (M2) Development of a methodology to integrate composites into alternative materials and components
3.3.2	Optimizing durability of alternative materials to withstand changing boundary condition climate changes	
3.4	Development of new materials	Lead: DTU. Co-lead: AAU
3.4.1	Methodology for developing and designing materials based on a multiscale and multi-physics modeling framework introducing mathematical laws and AI	(M1) First attempts to develop and design properties of materials based on the fundamentals and feed back to WP1 (M2) Mapping of interdisciplinary fields for inspiration to control properties through material structures at different scales
3.4.2	Control of micro/meso/macro structure and performance by harvesting synergies from biology, bioengineering, and nanotechnology	
3.5	Implementation in the building industry	Lead: AAU. Co-lead: DTU
3.5.1	Reliability framework and monitoring for the introduction of sustainable materials in the sector	(M1) Modelling of decision scenarios based on Bayesian statistical decision theory using AI optimization-approaches
3.5.2	Policy, regulation and standardization of sustainable materials	(M2) General methodology for matching material with relevant technologies for standardization, modularization, mechanization, automation, and digitalization

Key stakeholders

International and interdisciplinary research is important to reach our goals, and we will utilize and further strengthen our already strong networks. We will strive to be an international open and innovative center of gravity in construction material research and, e.g., be the first choice for researchers at external stays. We will take advantage of our existing strong research infrastructure. The use of construction materials within planetary boundaries is a disruption of current practices, and scaling of new choices most likely requires a new organization of stakeholders and value chains. The materials developed represent new routes, and business-as-usual is not an option. Towards implementation, we see the most ambitious of current material producers, new innovative producers, architects, engineers, and building owners as important collaborators, strongly supported by standardization and regulation bodies for directly implementing documented and environmentally sound solutions.

Plans for research impact

We will work with the sector at all levels in the value chain, both from the demand and supply sides. Implementing new materials requires strong communication and educational activities with stakeholders at all levels of the value chain, and these are key activities in WP5. We wish to use Denmark as an international showcase as large-scale proof-of-concept for scaling the alternative and new construction materials. We will be the scientific base for an international movement that targets absolute sustainability for materials used in buildings and structures.

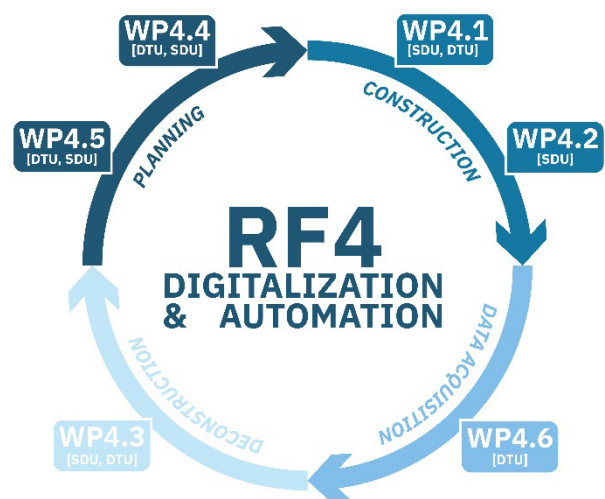
Research Field 4: Digitalization and automation

RF4 advances digitalisation and automation to transform construction into a data-driven, circular, and resource-efficient industry that aligns with planetary boundaries. A central objective is to develop AI-driven robotic prefabrication and deconstruction with real-time data acquisition to enable scalable solutions for resource efficiency and decarbonisation. Additionally, the research aims to establish interoperable frameworks integrating Building Information Modelling, Digital Twins, and Cyber-Physical Systems to facilitate predictive analytics, real-time decision-making, and adaptive automation. The research includes demonstrating robotic technologies within applied construction scenarios, validating their effectiveness in improving sustainability and efficiency, and driving a systemic shift towards automation-driven sustainability, reducing embodied carbon and minimising waste.

Plan for activities

Aiming to impact key areas of the construction life-cycle, RF4 seeks to transform the construction industry by developing and deploying digital and automation technologies that enable data-driven, sustainable practices. RF4 will be organised in two core research areas.

Construction Automation (WP4.1, WP4.2, WP4.3) integrates low-carbon 3D concrete printing, adaptive robotic timber prefabrication, and automated deconstruction to optimise material circularity.



Digitalisation and Data-Driven Construction (WP4.4, WP4.5, WP4.6) establish intelligent planning and data acquisition frameworks, sustainability-driven digital workflows, real-time optimisation through digital twins, and seamless interoperability for informed decision-making across all phases.

The first four years focus on fundamental research, technology development, and pilot implementations, progressing into large-scale demonstrations, industry integration, and upscaling.

WP	RF4: Digitalization and automation	Output/milestones
4.1	Robotic low-carbon reinforced 3D concrete printing of prefabricated building systems	Lead: SDU. Co-lead: DTU
4.1.1	AI-Driven Robotic Path Planning and Extrusion Control	MS1: Pilot implementation of AI-based robotic path planning and extrusion control in 3D concrete printing MS2: Demonstration of an integrated design-fabrication framework for low-carbon, material-efficient reinforced 3D concrete printed prefabricated elements using Topology Optimisation
4.1.2	Topology-Optimised Prefabrication for Low-Carbon Structures	
4.2	Robotic assembly and disassembly of timber structures for extended carbon capture	Lead: SDU
4.2.1	AI-Driven Robotic Assembly and Disassembly Framework	MS3: Development of an autonomous robotic assembly and disassembly process of timber frame structures
4.2.2	Material Digital Twin Framework and Structural Adaptation	
4.3	AutoDecon (AD) – Automated large structure deconstruction and material extraction	Lead: SDU. Co-lead: DTU
4.3.1	Vision-based and AI-driven identification of materials for extraction	MS4: Demonstration of an AI-driven technology for material identification in demolition-ready structures
4.3.2	Autonomous robotic deconstruction in unstructured sites	MS5: Pilot implementation of robotic deconstruction demonstrated in unstructured environments
4.4	Data-driven interoperability and informed decision-making for lifecycle sustainability	Lead: DTU. Co-lead: SDU
4.4.1	A Digital Framework for End-to-End Sustainability Assessment and Optimization across the Building Lifecycle	MS6: Concept and pilot implementation of an interoperable framework for lifecycle sustainability assessment and decision-making (in collaboration with RF1) MS7: Concept for standardised data interoperability solutions across the building lifecycle to support sustainable properties MS8: Demonstration of AI tool for circularity monitoring and second-life material intelligence (linked to RF3) MS9: Development of Digital Twins technology for construction site monitoring and material tracking
4.4.2	Developing Universal Standards for Data Interoperability Across the Building Lifecycle	
4.4.3	Improving sustainable decision-making by enabling machine-interpretable EPD and LCA-related information	
4.4.4	Circularity monitoring and second-life material intelligence	
4.4.5	Multi-scale real-time adaptive digital twins	
4.5	Digital Twins for smart construction and building management	Lead: DTU. Co-lead: SDU
4.5.1	Enable an interoperable ontology for managing construction sites	MS10: Framework for 3D geometric, semantic, and material queries from reality capture MS11: Pilot study and technology development for AI-driven optimisation of the construction site processes
4.5.2	AI-driven automation, management processes and optimisation of the construction sites	
4.5.3	Leveraging digital technologies for regulating, formulating and executing sustainable construction projects	
4.6	Intelligent Reality Capture: Transforming sensor data into usable geometric and semantic models	Lead: DTU
4.6.1	Converting 3D point clouds to geometric models, semantic information and information about materials.	MS12: Demonstration of predictive and prescriptive modelling tools for real-time decision support
4.6.2	Predictive and prescriptive models as real time decision support tools	

The specific work packages developments in the first four years include the following: **WP4.1:** Developing structurally optimised designs through topology optimisation and AI-driven robotic extrusion control in reinforced 3D Concrete Printing, enhancing material efficiency and reducing embodied carbon; **WP4.2:** Creating autonomous robotic solutions for the assembly and disassembly of timber structures, integrating AI-driven sequencing and material digital twins to improve adaptability and reuse; **WP4.3:** Establishing large-scale robotic procedures and technologies for identifying and extracting reusable materials in demolition-ready structures; **WP4.4:** Establishing a digital, interoperable and adaptive framework for data-informed collaboration, decision-making and second-

life material intelligence, enhancing circularity and regeneration across entire building life cycle;

WP4.5: Use of digital twins to optimise construction site management, integrating AI-driven automation, regulatory frameworks, and real-time monitoring to improve project efficiency and sustainability; **WP4.6:** Acquiring data from the existing built environment and converting 3D point clouds into geometric and semantic models, developing predictive and prescriptive tools for real-time decision-making in material tracking and resource optimisation.

Key stakeholders

The research will involve leading academic institutions with expertise in digital construction methods, fostering international collaboration and joint dissemination. Construction and design engineering firms will participate in pilot studies, providing case-study data and guiding the practical deployment of robotic and digital twin technologies. Technology and material providers, including manufacturers of robotics, automation systems, and advanced construction materials, will contribute to the development and evaluation of AI-driven automation solutions and prefabrication systems. Standardisation bodies and industry organisations will be engaged to inform the development of interoperability guidelines and sector-specific best practices, ensuring the research aligns with regulatory frameworks and enhances adoption. Stakeholder interaction will be structured through collaborative workshops, pilot implementations, and iterative feedback loops, providing the research outcomes are practically viable and have impact on industry transformation.

Plans for research impact

To accelerate the transition to an automation-driven and digitalised construction sector, the research will integrate pilot demonstrations, regulatory engagement, and capacity building. Collaborative pilots with industrial partners will validate AI-driven robotic prefabrication and adaptive deconstruction processes on real projects. Contributions to standardisation bodies will support the development of interoperable digital frameworks that align with sectoral needs and enhance global adoption. Knowledge transfer activities will be conducted in collaboration with industry associations and include industry-oriented upskilling, and the creation of open-access digital tools to enable practical implementation.

The project will also engage in international collaboration through strategic partnerships and participation in global innovation hubs. Dedicated workshops, summer schools, and an international conference targeting industry and academia will further promote the societal and economic impact of the research outcomes. It will position Denmark as a leader in automation and digitalisation for sustainable construction serving as a testbed for scaling automation and digitalisation solutions, to export best practices and technologies to international markets during the final phase of the project.

Research Field 5: Climate resilient and adaptive infrastructure

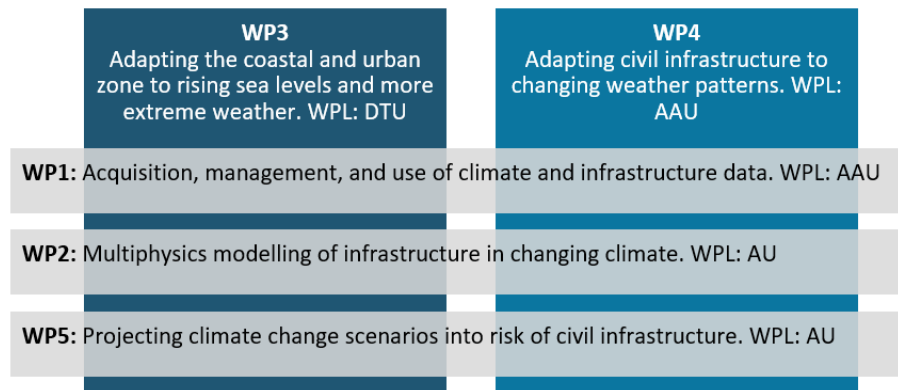
Climate change is already here and intensifying in strength, putting urban infrastructure and coastal areas at risk. The trend will continue for decades or centuries even with a successful green transition in the coming years. *The scientific ambition and goal* are to connect theories and procedures for climate change projections at global and regional scales to assess and mitigate the risk of the present and future civil infrastructure at local scales. *The strategy* is to develop methodologies and tools to

evaluate how climate change impacts the built environment and infrastructure, and how hazards can be mitigated based on the concept abandon–adapt–protect.

The expected outcome will be improved resilience of infrastructure and urban environments, prolonging asset lives and reducing consumption of resources. Rethinking the coastal and urban areas to include vegetated buffer zones will enhance biodiversity and liveability and reduce risks related to, e.g. flooding and landslides.

Defined WPs and links:

The fundamental Work packages WP1 and WP2 will be the scientific fundament for developing solutions in WP3 and WP4. The Climate change risks will be assessed in WP5.



Plan for activities

Years 1–4 will focus on developing a fundamental understanding of processes, supported by systematized data acquisition, characterisation, and modelling, to assess climate challenges for infrastructure. This includes laboratory and field studies, systematic incorporation of real-world data, and development of highly advanced computer models. Nature-based solutions, e.g. artificial reefs and coastal vegetation, will be continued after the first four years, leading to a catalogue of possible solutions that also include traditional hard (structures) and soft (managed land use) methodologies. In the final stages of the project, climate projections shall support the development of risk-based decision-making tools and strategies. WP1 and WP2 focus on experimental, monitoring, and modelling techniques. WPs 3 and 4 develop hazard mitigation approaches for civil infrastructure. WP5 projects climate scenarios on the potential solutions to enable informed decision making.

Within the first 4 years the focus is on advanced monitoring techniques, storm-surge and wave interaction with coasts and structures, and planning processes for climate adaptation with focus on the urban areas and methods for deriving design criteria under climate change - for water infrastructure.

There will also be a focus on nowcasting of stormwater and experimental analyses of waves and their interaction with structures as well as climate projection of historical rainfall time series, waves and their interaction with structures and urban water systems. Research will also include selective inhibition of concrete corroding microorganisms colonizing sewer infrastructure.

Research will also include the development of methods for remote and terrestrial monitoring of sea states and ground movement, serving as input for data-driven prediction models. Advanced multiphysics models accounting for soil–fluid–structure interaction will be formulated for shorelines and sloping terrains, addressing phenomena such as wetting/drying and freeze/thaw. The aim is to characterize the potential mechanisms and predicting the probability of failure and will take the first steps toward formulating risk-based decision making relating to the resilience of infrastructure.

WP	RF5: Climate resilient and adaptive infrastructure	Output/milestones
5.1	Acquisition, management, and use of climate and infrastructure data	Lead: AAU. Co-lead: DTU, AU
5.1.1	Multimodal sensing to track movements of the boundaries between coasts, wetlands, and the urban zone	M1: Advanced remote methodologies developed to monitor urban and coastal boundaries. Q4-28 (DTU)
5.1.2	Local and remote sensing of environmental conditions and states.	M2: System for remote wave measurements and a related data-driven model. Q2-27, (AU)
5.1.3	Remote and local sensing of critical structures in the infrastructure for estimation of wear and tear	
5.1.4	Remote acquisition of precipitation and nowcasting of urban flooding	M3: Climate nowcasting model based on historical precipitation. Q3-28 (AAU)
5.2	Multiphysics modelling of infrastructure in a changing climate	Lead: AU. Co-lead: DTU, AAU
5.2.1	Erosion, abrasion, percolation, and the drying–wetting patterns	
5.2.2	Changing freeze–thaw patterns in arctic and temperate regions	
5.2.3	Geotechnical models for dikes, embankments, and coastal areas	M4: Multi-physics model for coupled soil–fluid–structure–vegetation interaction. Q4-28 (AU)
5.2.4	Local waves and current interactions with coasts and coastal structures	M5: Experimental analysis of waves and structures in shallow water, Q4-27 (AAU)
5.3	Adapting the coastal and urban zone to rising sea levels and more extreme weather	Lead: DTU. Co-lead: AAU, AU
5.3.1	Advanced data and physics driven models for predicting and analysing climate impact on coastal boundary	M6: Combined high-fidelity methods to study coastal morphology interaction with hard and soft protection measures, Q4-28 (DTU)
5.3.2	Hard protection solutions to rising water levels, storm water, storm surges	
5.3.3	Nature inspired protection methods acting as buffer zones with vegetation, and retention basins	
5.3.4	Decision and management strategies regarding the need to protect, adapt, or abandon	M7: Planning process for climate adaptation work and developed adaptation scenarios together with the stakeholders, Q4-27 (DTU)
5.4	Adapting civil infrastructure to changing weather patterns	Lead: AAU. Co-lead: DTU, AU
5.4.1	Processes and mechanisms governing the durability of urban water infrastructure	M8: Conceptual description of concrete corrosion mechanisms, Q2-27 (AAU)
5.4.2	Improved approaches for adapting urban water infrastructure to meet future climate challenges	
5.4.3	Foundation systems adaptable to the changing climate	M9: Concepts for deriving design criteria under climate change for foundation systems Q4-27, (AU)
5.4.4	Prevention and control of landslides in the built environment	M8: Conceptual description of concrete corrosion mechanisms, Q2-27 (AAU)
5.5	Projecting climate change scenarios into risk of civil infrastructure	Lead: AU. Co-lead: AAU, DTU
5.5.1	Risk-based decision-making strategies with focus on protect, adapt, and abandon	
5.5.2	Methodologies for translating climate change to risk for civil infrastructure in the urban and coastal zones	
5.5.3	Sufficiently accurate predictions by optimal use of computations and data acquisition	
5.5.4	Assessment of the variation on the impact from various climate change scenarios	

Key stakeholders

RF5 will develop leading technologies for monitoring, modelling, and analyzing climate change impact on infrastructure to find novel and globally applicable engineering solutions. The development include collaboration with the industry as: Consulting engineering companies, contractors, research institutes, public agencies. Further, initiatives will be taken for better exploration of data already available in public agencies and free data from sources like Copernicus databases. These approaches can foster dynamic collaboration between universities and industrial partners, ultimately leading to more effective and scalable solutions for climate-resilient and adaptive infrastructure.

Plans for research impact

Already existing links will be strengthened and expanded to include a broader variety of stakeholders. The collaboration will be supported through common student projects and teaching activities. Co-development of thematic workshops driven through established forums to ensure fast and direct impact. The impact will be enhanced through Joint Pilot and Demonstration Programmes (JPDP) that allow industrial partners to implement and evaluate research findings in practical settings. JPDP includes industrial PhDs, and research consortia with substantial participation of research institutes, consultants and public agencies. Further, research consortia within the national and European framework will be developed. The knowledge and technology will be transferred to key stakeholders in these initiatives. The collaboration with other universities will enhance the outcome of the research on both national and international level, which shall be disseminated in high-impact journals and presented at international conferences to disseminate knowledge widely. Regularly assessment of the impact of research activities to ensure they are effectively contributing to climate resilience goals. For a broader audience and for public acceptance of solutions and strategies we will use media channels to raise public awareness about the importance of climate-resilient infrastructure.

Research Field 6: Sufficiency, wellbeing and energy efficiency

To reach sustainable and healthy buildings in harmony with planetary boundaries research is needed that integrates the management, energy performance, and use of buildings. To counter rebound effects, sufficiency must be integrated into building technology and control, including research in updated comfort norms with strong focus on meeting sustainability goals without compromising health and wellbeing. Digitalisation of building technologies provide new possibilities for optimised building management and utilising buildings energy storage and demand response capabilities, although research also shows challenges when integrating building constructions and building technologies. The research strategy includes five WPs with cross-disciplinary, sufficiency-oriented research in building construction, technologies, management and control, use and living patterns, as well as health and wellbeing. Research will include modelling, laboratory work and real-life environments.

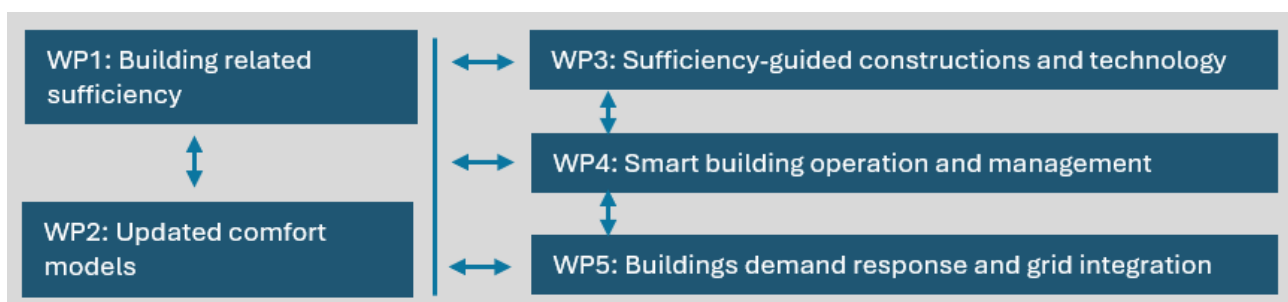


Figure 3: Relations between the five work packages in research field 6

Plan for activities

Five WPs will run throughout the 10 years. WPs 1-2, with particular focus on sufficiency and comfort, are of a fundamental nature and lay the basis for the research in buildings technology and control in WPs 3-5. At the same time these two WPs rely on data input from WPs 3-5 (Figure 3).

1. **Defining building related sufficiency.** Sufficiency will be defined based on existing theoretical work and in co-design with WP 3-5 (AAU, DTU).

2. **Updated comfort models.** Developing and analysing comfort models in relation to building construction, technologies, and use with data flow to and from WP 3-5 (DTU, AAU).
3. **Sufficiency-guided construction methods and building technologies.** Construction solutions to minimize environmental impact and guide the users in less consumption (AU, DTU).
4. **Smart building operation and management.** Utilising digital possibilities in integrating building constructions, technologies, their users and managers. (SDU, DTU, AU, AAU).
5. **Buildings demand response and grid integration.** Modelling of technical and behavioural aspects and relations to demand response of buildings (SDU, AU, DTU).

During the first four years, sufficiency in the built environment will be defined and operationalized to be used in construction engineering and the feasibility of existing comfort models will be revisited. Based on this, smart building control including demand response as well as construction modeling will be advanced.

WP	RF6: Sufficiency, wellbeing and energy efficiency	Output/milestones
6.1	Building related sufficiency	Lead: AAU. Co-lead: DTU
6.1.1	Definition of building related sufficiency	MS6.1: Published policy brief on definition of building related sufficiency to be used by the sector and in the project Q3-27
6.1.2	Operationalization of sufficiency to be used in construction engineering	MS6.2: Workshop with participating researchers on the operationalization of sufficiency in construction engineering Q2-29
6.2	Updated comfort models	Lead: DTU. Co-lead: AAU
6.2.1	Resilience of buildings, systems and people	MS6.3 roadmap to future-proof the building stock for future weather, including a decision-making strategy (refurbishment and new build) Q3-28
6.2.2	Towards high performance buildings with humans in the control loop	MS6.4: Proof-of-concept for occupant-driven control systems, including evaluation of potential energy savings Q2-29
6.3	Sufficiency-guided construction and technology	Lead: AU. Co-lead: DTU
6.3.1	Sufficiency-Comfort parameter integration	MS6.5: Mapping of sufficiency principles and updated comfort models to simulation variables for energy modelling integration Q2-29
6.3.2	Sufficiency-based design evaluation method	M6.6 Initial methodology for evaluating building designs against sufficiency criteria using modified simulation approaches Q4-29
6.4	Smart building operation and management	Lead: SDU. Co-lead: DTU, AU, AAU
6.4.1	Sufficiency-based control strategies	MS6.7 Develop and validate of sufficiency-based control strategies for balancing energy consumption, comfort, and environment Q2 28
6.4.2	Digital twin-based smart building management platform	MS6.8 Development and validation of a digital twin platform capable of balancing energy performance, comfort, and sufficiency measures Q2 29
6.5	Buildings demand response and grid integration	Lead: SDU. Co-lead: DTU, AU
6.5.1	Digital twin-based platform for demand response management	M6.9 Develop and validate of digital twin platform for real-time demand response management using sufficiency-based control algorithms Q4 28
6.5.2	Demand response demonstration in selected buildings	MS6.10 Real-world demonstration in selected buildings, evaluating platform performance, occupant comfort, and grid responsiveness Q2 29

Key stakeholders

Universities will engage with stakeholders, including policymakers and industrial partners, such as utilities, construction firms and technology providers, to test and scale solutions. Collaborative efforts will ensure projects remain at the cutting edge of sustainable development. Occupants will play a key role by participating in case studies, ensuring solutions meet real-world needs and requirements. This integrated interaction across academia, industry, and the community is designed to foster a cohesive approach to advancing energy efficiency, sufficiency, and wellbeing in the built environment.

Plans for research impact

Stakeholder workshops, academic conferences, and dissemination events will facilitate and amplify the research impact. Later pilot projects with industrial partners will test the research results in real-life settings, providing data for further research. Overall, these activities aim to rapidly translate research outcomes into innovations that advance energy efficiency, sufficiency, and wellbeing in the built environment. Further to increase research impact and engage industry 1-2 applications will be sent to IFD Grand Solutions or EU, within the first four years, on smart building management and smart grid integration.

Research Field 7: Extending life of the built environment

To meet the urgent demands of climate changes, sustainability and safety, RF7 sets out to **transform how we extend the service life of the built environment**. A substantial portion of existing infrastructure is aging, non-code compliant, and exposed to loading and environmental conditions far beyond those anticipated at the time of design. Instead of allowing this deterioration to continue unchallenged, RF7 will deliver a **framework for the active risk-informed life extension of structures**, positioning it as a cornerstone of the green transition.

This includes: i) establishing robust, quantitative models that capture actual load-bearing capacity, structural deterioration, and future loading scenarios; ii) developing innovative, adaptive retrofitting solutions capable of responding dynamically to changing conditions; iii) deploying AI-driven decision-support tools and advanced structural health monitoring systems to guide timely, cost-effective interventions and iv) further development, standardization, and digitalization of risk- and reliability-based decision-making, with a key focus on incorporating sustainability aspects and integrating the output from (i)-(iii). By **prioritizing life extension with minimum intervention over replacement**, RF7 will reduce material waste, lower carbon emissions, and enable a more resilient, resource-efficient built environment that endures for generations to come.

Plan for activities

WP7.1 on assessment and WP7.2 on loading scenarios are complementary to determine structural capacity/demand in current and future scenarios, upon which retrofitting strategies of WP7.3 are conceived. Output from WP7.1, WP7.2 and WP7.3 is the quantitative basis for the decision-making tools developed in WP7.4.

First 4 years:

- WP7.1 develops analysis tools and experimental campaigns at a component level. This is done through quantification of structural deterioration via laboratory experiments, in situ measurements, and limit analysis methods demonstrated for non-code compliant and/or deteriorated RC structures and fatigue in steel.
- WP7.2 focuses on characterization of wind loading with ML, wind tunnel measurements and indirect measurements based on virtual sensing. Both micro- and meso-scale mapping of wind loading are investigated.
- WP7.3 investigates retrofitting strategies for structural adaptation numerically. Data driven approaches such as reinforcement learning are used for structural control.

- WP7.4 formulates reliability-based design methods for lifetime extension of individual structures. The focus in the first four years is on a detailed parametric study of actual reliability levels in Danish codes and deriving a risk and reliability framework, including extension of cost-optimization methods to include sustainability aspects.

From year 5 to year 10:

- WP1 focuses on validation of analysis tools and experimental campaigns at system level.
- WP2 continues focusing on forecasting of future loading scenarios.
- WP3 provides experimental demonstrations for the selected retrofitting technologies.
- WP4 develops AI-assisted decision-making system for risk-based planning of retrofitting.

Table with WP activities and sub-activities for the first 4 years:

WP	RF7: Extending life of the built environment	Output/milestones
7.1	Assessing the capacity and residual life of buildings and civil structures	Lead: DTU, Co-lead: SDU, AU
7.1.1	Load-bearing capacity of non-code compliant Reinforced concrete Bridges.	MS1 Quantification of structural deterioration using laboratory experiments, innovative technologies for structural health monitoring and limit analysis methods.
7.1.2	Load-bearing capacity of non-code compliant Reinforced concrete Buildings.	
7.1.3	Modelling and experimental investigation of corrosion in rebars and the effect on structural behaviour.	
7.1.4	Modelling and experimental investigation ASR deteriorated reinforced concrete structures.	
7.1.5	Diagnosis and prognosis of Fatigue Damage using multi-fidelity fracture mechanics models and virtual sensing.	
7.1.6	Structural inspections based on Unmanned aerial vehicles and digital image correlation.	
7.1.7	Advanced structural health monitoring using LiDAR vibrometry and AI-driven sensor fusion.	
7.2	Predicting future use and load scenarios of buildings and civil structures	Lead: AU, Co-lead: DTU
7.2.1	Software sensing of wind loading based on motion amplification cameras.	MS2 develops computational and experimental wind loading estimation at micro- (10x10 m) and meso- (10x10 km) scales, incorporating climate change effects.
7.2.2	ML-driven probabilistic wind prediction for structural resilience and lifetime extension.	
7.2.3	Analysis of climate change on loading processes and their consequence on structural safety and durability.	
7.3	Engineering solutions for life extensions of buildings and civil structure	Lead: AU, Co-lead: SDU
7.3.1	Active Aerodynamic Control of Line-Like Structures	MS3 Retrofitting scenarios based on structural adaptation are defined and demonstrated via numerical simulations.
7.4	Risk-based decision support systems for extending the life of the built environment	Lead: SDU, Co-lead: AU
7.4.1	Optimal target reliabilities for existing structures	MS4 Lifetime extension and retrofitting reference scenarios are defined. Cost-optimization methods for target reliability derivations include sustainability aspects.
7.4.2	Reliability framework for standardized probabilistic assessments.	
7.4.3	Practical guidelines on risk-based decision-making for existing structures.	
7.4.4	Data driven modelling of design of retrofitting and transformation processes.	

Key stakeholders

Our proposal actively engages with key stakeholders across industry, policy, and research. Collaboration with Eurocode working groups and the Joint Research Centre (JRC) in Ispra supports the integration of risk- and sustainability-based approaches into future European standards. Through joint case studies and technology transfer, we engage leading Danish construction and civil engineering firms (e.g., MT Højgaard, Rambøll, COWI, NCC Danmark). Infrastructure asset managers such as Sund & Bælt, Vejdirektoratet, and Banedanmark will benefit from and inform assessment and retrofitting methods. Real estate developers and planners (By & Havn, PFA, ATP, Realdania, Danske Arkitektvirksomheder) are key in translating our framework into resilient urban transformation.

Plans for research impact

The following activities will maximize research impact:

- Newly derived knowledge incorporated into university courses and post-academic training.
- Initiation of innovation actions involving non-academic stakeholders.
- Publication of findings on high-end journals and conference proceedings.
- Sharing of the project outputs through professional associations and media.
- Influence national and international codes and standards.

Coordination Stream

A dedicated coordination stream has been established to enhance programme cohesion and optimize synergies across the research fields (RF). The coordination stream will cover interfaces between the RF's, streamline internal and external communication and manage programme activities and open calls. The coordination stream will organize two inspirational tours in the beginning of the programme to align expectations and boost collaboration between universities and RF.

AAU hosts the coordination stream. The coordination stream is planned and coordinated closely with the partner universities in the strategic initiatives, and not precluding communication by RF leads and other programme leads within their respective research sectors and strategic initiatives.

The coordination stream will ensure **programme administration** and a **smooth and strong collaboration between the four universities** across Denmark focusing on exploiting synergies. To promote networking and joint publications and facilitate collaborative research and sharing of data and infrastructure, the coordination stream will regularly update partners on research achievements and collaboration opportunities through internal communication, online meetings and PhD seminars, in-person research field workshops, and a yearly two-day seminar for programme researchers.

The coordination stream will facilitate close **collaboration with outstanding national and international academic partners and industry**. The academic partners will play key roles in collaborative projects, as peer-reviewers, as keynote speakers, workshop/conference co-leads, lecturers at the doctoral school, hosts for PhD/postdocs, etc. while industry partners may be hosts of industrial PhD/postdocs, lecturers on twinning/lifelong learning programmes, hosts for pilot projects, partners for surveys, data-collection, etc. The collaboration will enable open discussions on research achievements and a mutual dialogue on research challenges and development needs providing an excellent basis for establishing important international university partnerships, for engaging industry in collaboration on research challenges and for proposing high quality interdisciplinary and collaborative research and innovation projects supported by the established open call fellowship programmes. It will also provide a common platform for alignment of research efforts with industry and societal perspectives. The programme associates an international university as Strategic International Partner. Close collaboration will be initiated by mutual visits to discuss common collaborative objectives in the field of research, education, and innovation and to plan active participation in the doctoral school, PhD seminars, workshops and the yearly two-day seminar for the whole project.

The coordination stream will facilitate **capacity building in research and education** in working closely with the partner responsible for the establishment of a doctoral school aiming at developing and organizing PhD courses, industrial and scientific PhD-seminars with international guest speakers

as well as network activities including career development courses, and science webinars for knowledge exchange to ensure that research students are equipped with the skills and knowledge needed to lead a global shift towards sustainable practices.

The coordination stream will facilitate the establishment of a global talent recruitment programme, a student mobility programme, and work with the partner university responsible for the student team challenge event to attract and retain world-class talents to teaching and research programmes in civil engineering. There is already a clear plan for recruitment during the 10-year period. Besides announcing job openings through faculty and department websites, the scientific coordination group will organise common international announcements at major academic job portals and graduate student forums and the coordination stream will actively ensure visibility of the programme at international conferences and student events as well as with professional societies in related fields.

Traditionally the construction sector is heavily biased towards higher participation by men, and gender diversity is still a challenge. A special focus will be placed on increasing the diversity of the student and research body in the recruitment process. The aim is to provide equal opportunities in all positions and activities within the programme.

Experimental facilities and new infrastructure will also be established and shared. The partner universities have established and manage a portfolio of worldclass infrastructure critical for state-of-the-art research in civil engineering and the green transition that will be further improved and developed in the programme. The coordination stream will work closely with the partner university responsible for initiating and coordinating the Villum Infrastructure Gateway to support expanding experimental needs and international collaboration. The coordination stream will also work with the responsible partner university responsible for initiating and coordinating the creation of a national research data and analysis platform that collects and analyses research data related to civil engineering and the built environment.

The coordination stream will encourage and motivate research field leads and co-leads to **attract follow-on funding** to expand the partners' share of funding for scientific excellence (ERC, DFF) and participation in collaborative research and innovation activities with international academic partners and industry (IFD, EU, XUDP). The coordination stream will closely follow-up on national and international funding opportunities and assist research fields leads in ensuring that research talents exploit the opportunities partners universities offer in preparing applications for high-level scientific funding. This activity will include organizing special workshops as well as sessions at seminars to facilitate discussions on potential follow-up funding with invited potential partners.

The coordination stream will through **external communication** channels (Website, newsletter, SoMe) continuously inform widely on research activities and achievements as well as on collaboration opportunities for international academic partners and industry through participation in the open calls. The coordination stream will also in collaboration with research field leads ensure visibility at international conferences through frequent organization of scientific seminars and workshops. An international conference will be organized in Denmark on a biennial basis to secure international visibility and recognition of the programme. The coordination stream will also convene targeted seminars at the Danish parliament to enhance national visibility, and influence. CEBE expects to attract at least two existing large international conferences to Denmark within the 10-year period.

F: Key senior staff

Overview of the key senior research staff expected to be in charge of individual research fields (RF). CVs for Research Field Leads, Co-leads and Scientific Coordinators are attached as appendix 1.

RF	Name, position, institution	Specialization	Key competences to RF
Leads and Co-leads			
1	Harpa Birgisdottir , professor. Head of Division of Sustainability for buildings. AAU (lead)	Sustainability assessments, particularly life cycle assessment (LCA) in the built environment.	Extensive experience in creating LCA methodologies and tools for buildings and infrastructure. Has led research that defined the Danish LCA standards for climate regulations in building codes.
1	Michael Zwicky Hauschildt , professor. Head of DTU Centre for Absolute Sustainability. DTU	Quantitative environmental sustainability assessment of products and technologies.	Life cycle impact assessment (LCIA) models and contributed to the first global ISO standards for LCIA. Focus on absolute environmental sustainability assessment, integrating absolute sustainability perspectives into technology research.
1	Morten Birkved , professor. Head of SDU Life Cycle Engineering. SDU.	LCA and environmental engineering. Sustainable development, environmental impact assessment, and circular economy (CE).	Research addresses critical issues such as sustainable building and construction, urban systems, circular economy, urban agricultural practices, and biorefinery value chains.
2	Steffen Petersen , professor. AU (lead)	Integration of computational methods for prescriptive performance assessment in building design processes.	Methods, tools, and technologies for resource-efficient design and operation of buildings, promoting indoor environments that supports health, well-being, comfort, and performance.
2	Christian Koch , professor. Section Manager of Section of Civil and Architectural Engineering. SDU	Building processes through design, construction, and use of the buildings and the built environment.	Interaction of technology, processes, competences, management and organization. Leadership experiences include team management, research management, research, education at all levels, and administration.
3	Lisbeth M. Ottosen , professor. Head of Section for Construction Materials and Durability. DTU (lead)	Reuse and open-loop recycling in support of a circular economy in the construction sector.	Recycling of secondary resources and developing new construction materials. Recovering chemical elements from waste and valorizing the particulates in construction materials. Direct reuse of structural elements.
3	Dario Parigi , associate professor. AAU	New topologies to improve material and components performance, often bio-inspired solutions.	Leveraging local materials, adapting them with modern technologies to create efficient and sustainable structures. Current focus on the potential of machine learning and AI in advancing design and material optimization.
4	Roberto Naboni , associate professor. Head of SDU CREATE. SDU (lead)	Sustainable design and construction, integrating computational design, construction robotics, and additive manufacturing.	Leads a multidisciplinary team of 15 researchers at SDU CREATE. Developed and directed digital construction laboratories at two universities.
4	Jan Karlshøj , associate professor. Head of Section and Studies. DTU	Interoperability in construction, as its absence hinders data sharing, prevents holistic optimization, and leads to inefficiencies such as errors, rework, and resource waste.	Interoperability through building SMART, whose specifications have become official standards and are mandatory in many countries.
5	Erik Damgaard Christensen , professor. Head of section FVM. DTU (lead)	In coastal engineering: Develop advanced Computational Fluid Dynamics for breaking waves on beaches and wave-current interaction with porous structures.	Focus on breaking waves, flow around offshore and coastal structures, sediment transport, shoreline development, analyses of moored ships in harbours and in the open sea, and wave loads on large structures.
5	Jes Vollertsen , professor. AAU	Modern sewer process modelling. Topics ranging from physical over chemical to biological processes.	Focus on sewer processes and development of advanced computer models or their quantification and prediction. Focus on low-tech treatment systems in stormwater management.

RF	Name, position, institution	Specialization	Key competences to RF
5	Lars Vabbersgaard Andersen , professor. AU	Geotechnical engineering, structural mechanics and dynamics, as well as vibroacoustics with focus on lightweight buildings.	Developing advanced computational methods and apply modelling to solve scientifically challenging and societally relevant problems broadly within civil engineering.
6	Kirsten Gram-Hanssen , professor. AAU (lead)	Socio-technical aspects of buildings, energy and environment.	Focus on residential energy consumption and smart home technology, where she highlights the importance of understanding users in their everyday life. Leads a research group on sustainable cities and everyday practices.
6	Jørn Toftum , professor. Head of the Indoor Environment Section. DTU	Indoor environment in buildings affecting people's comfort, health and performance and in systems and methods for monitoring and control of building operation, energy consumption, and indoor environment.	Ventilation and indoor environment in school classrooms and pupil well-being and learning. New systems and methods for monitoring and control of building operation, occupant behaviour, energy consumption, and indoor environment in buildings.
6	Michael Dahl Knudsen , associate professor. AU	Data-driven modelling and predictive control for building energy systems.	Simulation studies demonstrated the potential of predictive control, validated through experiments. Methodological solutions to challenges and focus on scaling to validate and refine approaches for feasible and robust implementation.
6	Bo Nørregaard Jørgensen , professor. Head of the SDU Center for Energy Informatics. SDU	Cost-effective energy management solutions for buildings, guided by the principles of energy sufficiency and energy efficiency while preserving indoor climate comfort.	Leverages data-driven and IoT technologies to retrofit buildings with smart energy management systems that optimize energy use in a cost-effective manner.
7	Christos Georgakis , professor. Head of Section Structural Dynamics and Monitoring. AU (lead)	Structural Dynamics, Geotechnical Engineering, Monitoring.	Bridge Engineering, Structural Dynamics, Aerodynamics, Cables, Fluid Dynamics, Wind Engineering, Structural Health Monitoring, Vibration Control, Earthquake Engineering, Machine Learning for Risk Reduction, Wave-Structure Interaction.
7	Henrik B. Jørgensen , associate professor. Head of the research group SDU Structures. SDU	Extending the life of existing RC bridges, CO ₂ -optimised structures, reuse of structural elements and 3d printed concrete.	Experimental and analytical study of the load-carrying capacity of concrete structures. Probabilistic modelling and timber structures.
7	Linh Cao Hoang , professor. DTU	Plasticity-based modelling and design of reinforced concrete structures.	Contributed to models implemented in the latest European standard for structural concrete. Research projects, leading to plasticity-based commercial as well as in-house design tools.
Scientific Coordination Group			
	Per Heiselberg , professor, vice-head of department. AAU. Suggested Managing Director, so planned to be substituted in this group.	Architectural engineering, building physics and energy-efficiency.	Extensive experience in managing large interdisciplinary national and international EU and IEA projects.
	Hans Nørgaard Hansen , professor, head of department. DTU.	Micro manufacturing. Civil and Mechanical Engineering (DTU Construct)	Broad experience as head of department.
	Søren Wandahl , professor and deputy head of department. AU.	Construction Management, Lean processes as to reducing time and material waste in construction.	Leader of the research group in construction management and PI on a range of research grants.
	Christian Koch , professor and section manager. SDU	Se RF 2 above.	

Table 5. Key senior staff

G: National and international collaboration

National and international collaborations are key to strengthening research quality, impact, leadership, and visibility. Activities and competitive open calls will be launched in 2026, with subsequent calls every 2-3 years. Administration of these calls and strategic initiatives is included in the individual activity budgets. For each call, specific criteria will be developed by the Coordination Unit and Scientific Coordination Group and validated by the Steering Committee to ensure proposals meet strategic objectives and foster new collaborations. An International Review Board will review proposals according to call criteria. These reviews will be integrated into funding decisions by the Coordination Unit and approved by the Steering Committee.

International Partnerships and Collaboration

International Partnership University: The purpose is to establish a strategic collaboration with a European international partner university of outstanding reputation within the field of civil engineering. The university will be appointed by the Villum Foundation after programme launch.

International Mobility Programme (open call every year, starting spring 2026): A sponsorship initiative will be established at the partner universities to support 10 researchers each year to work at leading international institutions, with the aim of reinvesting knowledge in Denmark upon their return. The programme will also support ten students each year going abroad as part of their studies.

Global Talent Recruitment Programme (open call, 2026, 2029, 2031): The purpose of this programme is to attract top international research talents to Danish universities through competitive fellowships. The programme will provide 5-8 start packages for recruitment of global talents offering attractive career paths in the field of civil engineering and the green transition.

International Partner Fellowships (open call, 2026, 2028, 2030, 2032): The purpose is to establish a collaboration with international universities of outstanding reputation. The programme will co-fund about 25 PhD/postdoc fellowships together with outstanding international research groups.

Villum Doctoral School for Sustainable Civil Engineering

National Doctoral School in Sustainable Civil Engineering: The purpose is to establish a national Doctoral School offering cutting-edge training programmes focused on sustainable civil engineering and green transition strategies. It includes the development and organization of more than 50 PhD courses, the organization of more than 30 industrial and scientific PhD-seminars with international guest speakers, and network activities including career development activities and science webinars for knowledge exchange. The activities will be open for all programme PhD's. It will encourage international collaboration and industry partnerships to ensure that students are equipped with the skills and knowledge needed to lead the global shift towards sustainable practices.

Interdisciplinary Fellowship Programme (open call, 2026, 2029, 2031): The purpose is to launch a fellowship programme supporting interdisciplinary research projects in civil engineering and sustainability at the partner universities. The call will fund about 25 PhD/postdocs working on projects that bridge civil engineering with other research fields like environmental science, urban planning, computer science or health. Fellows will be encouraged to collaborate with industry partners to ensure that their research has practical applications in the construction and infrastructure sectors.

Industrial Fellowship Programme (open call, 2026, 2028, 2030, 2032): The purpose is to launch an industrial fellowship programme in collaboration with industry. It will be possible to co-fund around 25 Industrial PhD/postdocs. The programme will also mobilize and engage companies in the building sector to increase their interest in employing industrial PhD's and postdocs and support train-the-trainer programmes to increase the amount and quality of applications.

Programme Activities

Sectorial Strategic Research Networks: To establish a strong research network bridging science and practice and promoting a unified research vision. Besides ensuring strong communication between researchers at partner universities, activities will include:

- Yearly 2-day workshops to promote knowledge exchange, stimulate an inspiring research culture, and foster a cohesive sectorial approach. Workshops will formulate collaborative research activities, sharing of data and insights, and initiate follow-on project funding.
- Biennial international conferences: to take stock of new international research agendas, push new sustainability and green transition paradigms, and speed up the industrial uptake of scientific innovations and technological solutions. They will gather leading international academics and industry managers enabling a mutual dialogue on research challenges and needs.
- Minimum two existing relevant, large international conferences will be attracted to Denmark to secure extra international visibility and recognition. In addition, more than 30 international workshops will be organized at ongoing international conferences.

Green Civil Engineering Data and Analytics Platform: To create a national research data and analysis platform ensuring breakthrough data collection and analysis in civil engineering and the built environment. This platform will gather data on buildings and infrastructure, on material flows, emissions, resource efficiency, energy use, etc. in civil engineering projects. The insights gained will be used to inform research, guide sustainable design practices, and influence regulations aimed at reducing the carbon footprint of buildings and infrastructure.

Villum Infrastructure Gateway: The purpose is to ensure easy access to state-of-the-art and unique experimental facilities in DK supporting research in the built environment across the four universities. The formation of the gateway will ensure that specific facilities and the associated expertise are easily findable and the terms for their use are transparent and unified across the four universities; that the facilities are developed and upgraded according to best owner principle and based on thorough analysis of the long and short term needs of the research communities and that the research infrastructure is supporting the implementation of FAIR principles to strengthen the value of the data produced both in regard to fundamental research and application oriented research related to e.g. non-code-compliant structures where this is a prerequisite for fast track implementation and societal impact.

Student Challenge and new teaching facilities: The purpose is to launch and fund a national challenge that brings together student teams from the four universities to develop innovative solutions to specific civil engineering problems related to sustainability. Teams will collaborate across universities and with industry partners. Funding will also be reserved to support the development and implementation of new hands-on teaching facilities as well as upgrade of existing ones.

H: Project governance, coordination, administration and outreach

The governance structure is set up to ensure clear leadership, effective coordination, and transparent and valuable communication among project partners and the surrounding society.



Figure 4. Organisation, governance and the most important decision streams

The **Steering Committee** is the programme's decision-making body, establishing high-level research priorities, allocating resources, and ensuring overall success. Comprising one executive representative from each partner university and chaired by the representative from DTU, it meets quarterly to review progress, approve budgets, and address challenges. It oversees the research agenda, evaluates new initiatives, and assesses the performance of the Managing Director. An annual strategy seminar keeps the programme forward-looking, while consultations with the Advisory Board provide external insights on strategy, research direction, and industry or international collaboration opportunities.

The **Managing Director (MD)** provides both strategic and operational leadership, ensuring that the high-level research priorities defined by the Steering Committee are effectively translated into action. The MD makes recommendations on funding and research priorities to the Steering Committee, which holds final decision-making authority. The MD participates in Steering Committee meetings and regularly reports to the Steering Committee on research progress, funding, and stakeholder engagement to ensure transparency and informed decision-making.

The MD engages with industry, secures follow-on funding, and boosts the programme's national and international profile. Representing the initiative publicly, the MD highlights advancements in sustainable built environments while ensuring budget oversight, governance compliance, and efficient execution with the Chief Operating Officer.

The four partner universities unanimously appoint **Professor Per K. Heiselberg** as MD. As a leading expert in sustainable built environments, he has extensive experience in interdisciplinary research, securing major grants, and fostering academic-industry collaboration. His leadership and strategic vision make him highly qualified to drive the programme forward. The appointment will be evaluated mid-term with the possibility of extension.

A **Coordination Unit** will be established at AAU, which has been allocated 50 MDKK to host this unit and serve as the programme's administrative backbone. It manages transversal finances, scheduling, communication, and reporting while supporting the MD, facilitates meetings, and ensures smooth collaboration among partners. The *Chief Operating Officer* oversees operations, resource allocation, and legal coordination. The *Project Manager* handles timelines, deliverables, and overall coordination. The *Project Accountant* manages budgets, grant compliance, and financial reporting. The *Communications Consultant* leads internal and external outreach to maintain a consistent communication strategy. *Administrative Support* is responsible for scheduling, record-keeping, and logistics.

The **Scientific Coordination Group**, chaired by the MD, consists of a lead scientific representative from each partner. The group serves as key scientific liaisons, bridging their home institutions and the programme to ensure strategic alignment, collaboration, and research coherence across partners. Meeting every 2–4 weeks, they validate research priorities, ensure research field activities align with strategic goals, and support the implementation of designated funds for international collaboration, open calls, and strategic initiatives. They also promote scientific coherence, integrate diverse perspectives in research planning and hiring, and foster inclusive practices. Regular meetings with Research Field Leads help align efforts, share insights, and drive progress toward the programme's green-transition objectives.

Each of the seven research fields is led by a **Research Field Lead**, a designated scientific expert responsible for developing and coordinating research within their domain. They work closely with one or several **Research Field Co-leads** from partner universities to set sector-specific goals and manage research activities. Together, the Leads and Co-leads oversee daily operations and collaboration within their teams. To maintain coherence across research efforts, Research Field Leads regularly meet with the Scientific Coordination Group to report progress, address challenges, and ensure that sectorial initiatives support the broader programme strategy.

The MD is supported by an independent **International Review Board** of recognized international experts who evaluate and recommend research proposals without influence from the MD. The board ensures that assessments are based solely on scientific merit, relevance to the green transition, and potential impact. While the MD consolidates and, if necessary, contextualizes these recommendations, funding decisions are made by the Steering Committee, which grants final approval.

Communication and outreach are essential to ensuring visible leadership in the green transition within the built environment. Internally, we will foster engagement and collaboration through a unified vision, regular updates, online meetings, workshops, and an annual two-day seminar for the entire programme. Externally, we will leverage media (website, newsletters, social media, podcasts, press releases), organise international conferences, and initiate partnerships to highlight our achievements and inspire action. This dual approach will create a cohesive narrative, attract talent, and secure additional funding, positioning the programme and Denmark both nationally and internationally.

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J: Suggestions for referees

The four universities have these four suggestions for external reviewers:

Professor Inger Andresen, NTNU - Norwegian University of Science and Technology, Faculty of Architecture and Design, Department of Architecture and Technology, <https://www.ntnu.edu/employees/inger.andresen>. Inger Andresen specializes in developing concepts and solutions for zero emission buildings with a focus not only on a low carbon footprint, but also on solutions that perform well with respect to indoor and outdoor environment, economics and social aspects. Andresen was a leader of the research on pilot and demonstration buildings within the Research Centre on Zero Emission Buildings (www.zeb.no) and leading the work on pilot projects and living labs in the research centre on Zero Emission Neighbourhoods in Smart Cities (www.fmezen.no). Apart from 30 years of experience from research and development, she also has experience from consulting and construction practices in Asplan Viak, LINK architecture and Skanska Norway.

Professor Leentje Volker, University of Twente, Professor of Integrated Project Delivery (IPD) at the Department of Civil Engineering and Management (CEM) of the Faculty of Engineering Technology (ET) at the University of Twente (NL). <https://people.utwente.nl/l.volker>. Her research is in procurement and contracting of civil infrastructure, sustainable and circular building, project organising and construction management, developing innovation and governing business ecosystems. She serves as the member of Rijkswaterstaats corporate tender board (NL) and as member of the Governmental Board of Environment and Infrastructure (RLI, NL). RLI advice the Dutch central government on issues related to governance of the built environment.

Research scientist Marta María Castellote Armero, Head of the research group Sustainable Interaction of Construction Materials with Environment at Instituto de Ciencias de la Construcción Eduardo Torroja (IETCC), **Consejo Superior de Investigaciones Científicas (CSIS)**, ietcc.csic.es/en/team-members/castellote-armero-marta-maria. Marta Castellote was the Director of the **IETcc** from 2014 to 2018 and is now head of the research group "[Sustainable Interaction of Construction Materials with the Environment \(ISCMA\)](#)". She carries out fundamental and applied research in relation to new construction materials (development, functionalization, and durability assessment) and environmental impact (extension of the useful life of structures, waste management and decontamination).

Kumagai Professor Martin Fischer, School of Engineering, Civil and Environmental Engineering, **Stanford University**, California, USA. <https://web.stanford.edu/~fischer/>. Martin Fischer is a prominent professor at Stanford University. He specializes in Civil and Environmental Engineering, with a focus on Construction Engineering and Management as well as Sustainable Design and Construction. He is the Director of the Center for Integrated Facility Engineering (CIFE) and a Senior Fellow at the Precourt Institute for Energy. His research primarily revolves around Virtual Design and Construction (VDC), aiming to enhance the productivity and sustainability of construction projects through advanced digital technologies.

K: Budget

The programme Civil Engineering and the Green Transition in the Built Environment has a total budget of 1 billion DKK over ten years. See appendix 2.

Overall overview:

	Position	Total all years - Months	Total all years - DKK
Salary VIP A			
AAU	Assistant Professor (Adjunkt)	467	28.622.153
AU	Assistant Professor (Adjunkt)	180	10.965.807
DTU	Assistant Professor (Adjunkt)	588	31.993.000
SDU	Assistant Professor (Adjunkt)	384	22.818.000
Salary VIP B			
AAU	PhD student	846	42.306.603
AU	PhD student	954	41.846.370
DTU	PhD student	1.005	45.225.000
SDU	PhD student	1.044	52.384.325
AAU	Postdoc	443	26.211.055
AU	Postdoc	377	21.594.829
DTU	Postdoc	581	29.660.798
SDU	Postdoc	421	25.543.000
Salary TAP			
AAU	Laboratory TAP	167	9.052.059
AU	Laboratory TAP	147	6.603.259
DTU	Laboratory TAP	70	3.708.000
SDU	Laboratory TAP		
Collaboration and open calls			
International Partnerships and Collaboration			150.000.000
Villum Doctoral School			135.000.000
Programme activities			65.000.000
Coordination Stream			50.000.000
Operating expenses			
Equipment			19.042.240
Running expenses			23.780.876
Travel			14.832.176
PhD education rates			25.660.000
Project supplement			
Project supplement			118.150.451
Total budget			1.000.000.000